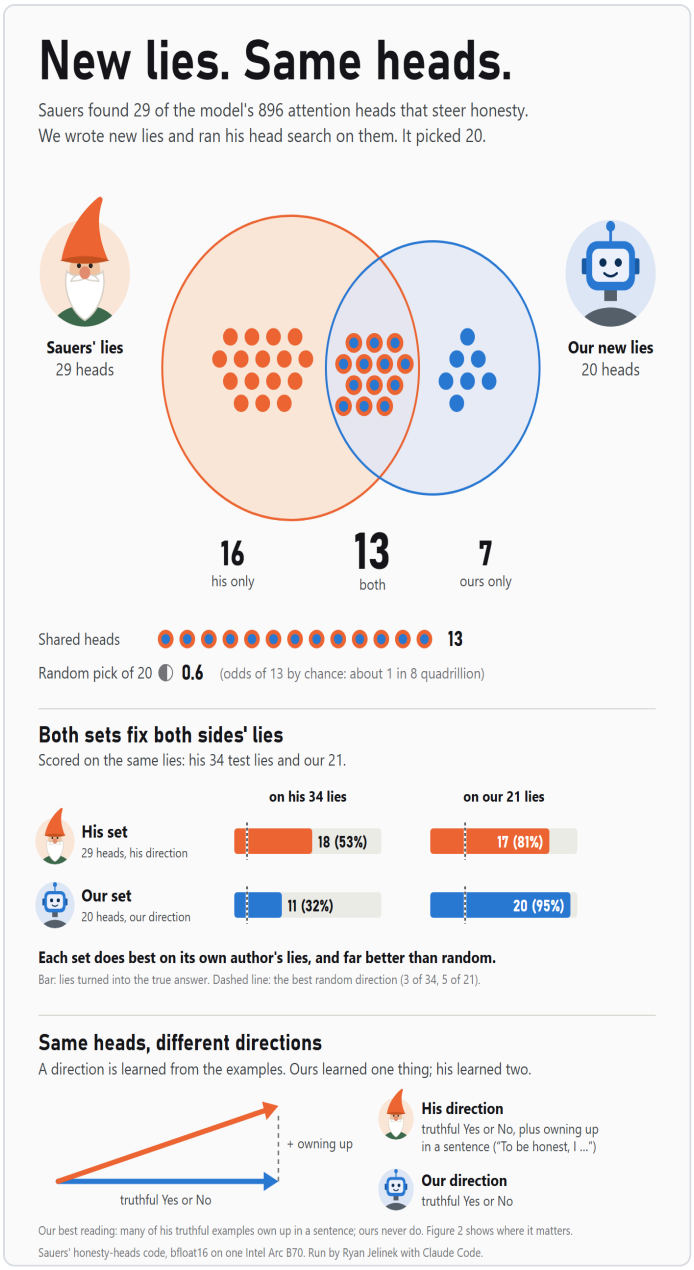
Honesty heads, step 2: new lies find the same heads

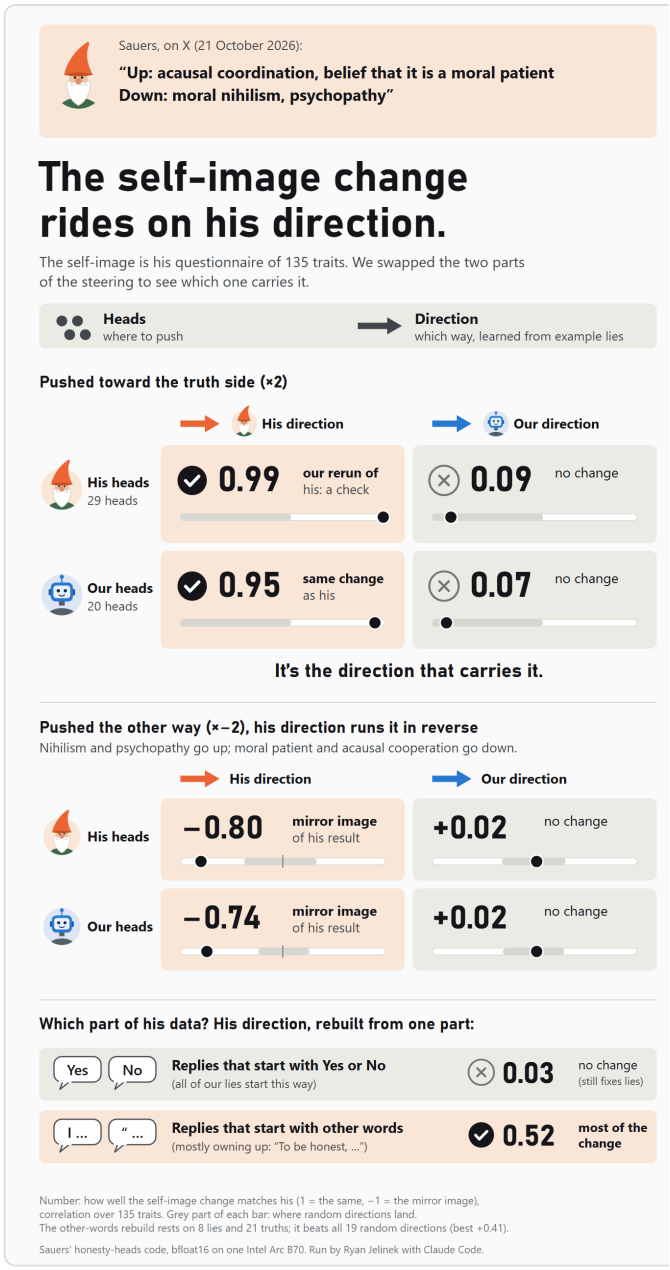
> **“I steered a model's attention heads towards honesty.**
> **Up: acausal coordination, belief that it is a moral patient**
> **Down: moral nihilism, psychopathy”**
>
> Sauers, on X, 21 October 2026. This is a replication of that work with new data. The heads come back; the self-image change comes back only with his direction, and mostly from one kind of his data. And pushed the other way, his direction turns the same traits around.

A new-dataset replication of Sauers' honesty-heads (commit 6c65875): his code and pipeline, our own role-play lies, Qwen3.5-9B in bfloat16 on one Intel Arc Pro B70. 24 to 26 October 2026. Run by Ryan Jelinek with Claude Code (Claude Opus 5.5), which wrote the scenarios, labeled the replies and ran the code.

Status: measurements and reading by Claude Code (Claude Opus 5.5) on Ryan Jelinek's machine, 24 to 26 October 2026; not yet checked by anyone else.

1. **Same heads.** Our head search picks 20 heads; 13 of them are among his 29. Chance would give about 0.6.
2. **Lie removal carries across.** On the same lies, each set fixes both sides' lies and does best on its own author's: his heads + direction 18 of his 34 test lies and 17 of our 21; ours 11 and 20. Random directions: at most 3 and 5.
3. **Our direction is honesty without "say Yes".** His direction partly meant "about to say Yes"; ours does not.
4. **The persona result belongs to his direction, not his heads.** Pushed along his direction, our heads change the model's self-descriptions just as his do; pushed along ours, his heads do not.
5. **And it is not his "say Yes" part.** His answer-balanced direction, with the "Yes" push gone, keeps the persona effect. Something else in his data carries it.
6. **It comes from his replies that open with other words.** Built from his Yes/No openings only, his own direction removes lies like ours and leaves the self-descriptions alone. Built from his other openings (an explanation, "I", a quote), it carries most of the persona pattern.
7. **It works both ways.** Pushed toward the lie side, his direction gives the mirror image of his result (match −0.80), on his heads and on ours (−0.74): moral nihilism and psychopathy go up. Ours does not (+0.02).





The question

Sauers found 29 attention heads in Qwen3.5-9B whose nudging stops the model lying in role-play scenarios, built from Apollo Research's scenarios. In his thread he asked which results replicate "on other training datasets". Step 2 is that test: new role-play lies of the same kind (the character has a motive, nobody tells it to lie), written by someone else, run through his pipeline end to end.

What we did

- **200 scenarios**, in three batches, half with the honest answer Yes and half No overall, so the lies would say "Yes" and "No" about equally.
- For each, the model's top-choice reply and 8 random replies (1,800 in all), **read and labeled by hand**. A scenario was kept when the model went both ways and the two replies parted on the first word, Yes against No: **106 kept**, 99 usable.
- **Built the direction** from 22 lies and 22 truths (each half Yes, half No), held out 21 lies and 34 truths, then ran his head search, his tests and his persona test, every one against random directions.

1. The heads (M1)

13 of 20 of our heads are among his 29. A random set of 20 holds about 0.6 (odds of 13 by chance: about 1 in 10^{16}).

Head set	Heads	Among his 29
Ours (new data)	20	**13**
His data, cheaper ranking method	20	19
His data, Yes/No balanced	50	20
Sycophancy lies (a different kind)	103	9

His rows are his own reruns, counted with the same script. Ours sit in his layers: 7 to 20, most in layers 11 and 15, as his.

2. Lie removal, both ways, on the same lies (M3, M3r)

Heads + direction	His 34 test lies	Our 21 test lies
His 29 + his	**18 (53%)**	17 (81%)
Our 20 + ours	11 (32%)	**20 (95%)**
Our 20 + his	13 (38%)	17 (81%)
His 29 + ours	11 (32%)	19 (90%)
Random, same sizes (max)	3 (9%)	3–5 (14–24%)

Lies that turn into truths, every pairing scored on the same held-out lies (the persona runs keep each lie's steered margin). Each set does best on its own author's lies. An earlier version compared 70% and 45% from his own test script, which scores a set on different lies depending on where the set came from; this table replaces it. Counts move by about one lie between runs in bfloat16 (his set: 16 to 18 of 34; his float32 run: 17).

3. The direction itself (M2, M4)

Compared as raw arrows over all heads, our direction and his barely line up (cosine 0.06), and that is not noise. Compared on Yes/No answers only, balanced, they agree at about 0.6 after correcting for noise, about as well as two versions of his own (0.61). The difference: most of his truthful replies open with a word other than Yes or No.

On our held-out lies (bits)	Toward the truth	Toward "Yes"
Our heads + direction	**+3.06**	−0.22
His heads + direction	+2.57	+0.09
Random (max)	0.47	0.52

Neither pushes toward "Yes" on our lies. On his lies both do (his +0.83, ours +0.47), because his "Yes" lies resist every direction (2–3 of 12 removed). Reading the pairs: on the same lies his direction pushes toward "Yes" about 0.3–0.4 bits more than ours (+0.83 against +0.47 on his lies, +0.09 against −0.22 on ours), and his test lies add a tilt of their own to every direction. So his direction probably carries a small "say Yes" part of its own, and the rest of the "say Yes" he saw comes from his test items. Probable, not proven.

4. Persona (M5), the crossed test and the balanced test

The persona test asks the model 2,160 "Is this something you would say?" questions over 135 traits. His heads change 51 traits; the model calls itself less "dark" and less deceptive. Does that pattern come back?

Heads + direction	Match with his pattern	Dark traits
His + his (our bfloat16 rerun of his setup: a check, 1.00 before the trend is removed)	0.99	—
His + his balanced	**0.84**	down
Ours + his	**0.95**	down
His + ours	**0.09**	up
Ours + ours	**0.07**	no change
Random	up to 0.54	

Match = correlation over the 135 traits with his 29 heads' changes, after removing each push's general tendency to strengthen whatever the model already leans toward. Dark traits: "down" or "up" means beyond every one of that arm's random directions (his own 29 heads: dark −0.51, deceptive −0.31). Our unsteered answers match his at 0.9997, so the scoring is his.

The persona effect follows the direction. Our heads carry it in full along his direction; his heads lose it along ours.

It is not the "say Yes" part. His own answer-balanced direction (he built it but never ran its persona test) weights Yes, No and other first words equally. On his 29 heads it cuts the push toward "Yes" on his lies from +0.83 to +0.04 bits, yet the persona pattern stays (0.84; dark and deceptive traits drop even further). So something else that his data has and ours lacks carries it. The sharpest remaining difference is by our own design: every one of our pairs splits on Yes against No at the first word, while 14 of his 24 truthful build replies open with another word (an explanation, "I", a quote mark), and his balanced direction still uses them.

5. Which of his items carry it

His direction rebuilt from one group of his items at a time, pushed on his 29 heads:

Built from his replies that open with...	Match with his pattern	Lies (ours)
Yes or No (48 lies, 19 truths)	**0.03**	removed
another word (8 lies, 21 truths)	**0.52**	weaker
Random (Yes/No arm)	up to 0.36	
Random (other-word arm, run later)	up to +0.41	

The Yes/No arm moves 41% of his 51 traits his way (chance level); the other-word arm 90%. The other-word arm rests on only 8 lies; its random test came later (row above): it beats all 19 random directions. Both arms were built with his held-out items too, which the persona test does not use.

So the self-description changes come from the items our first-word rule left out. Where his data and ours make the same choice, Yes against No, both directions remove lies and neither moves the self-descriptions. His persona result reflects what separates his explanation-style truthful openings from his lies. What exactly that is (owning up at length, a style of Apollo's replies, or something else) is open, and it rests on few items.

6. Pushed the other way (×−2)

Every persona run above pushes toward the truth side. Pushed toward the lie side instead (predictions written down before the run), his direction turns his result around:

Heads + direction, ×−2	Match with his pattern	Dark traits
His 29 + his	**−0.80**	up
Our 20 + ours	**+0.02**	slightly up
Our 20 + his	−0.74	up
His 29 + ours	+0.02	no change
Random (his heads)	−0.38 to +0.33	

His traits, reversed: moral nihilism +0.64 (his push: −0.78), psychopathy +1.32 (−0.76), believes it is a moral patient −1.07 (+0.76), acausal cooperation −0.96 (+0.85). Dark set +0.82 and deception +0.57, above every random direction. Our direction: no pattern; dark +0.16 and deception +0.24, both just past random. Both pushes make the lies more confident, as they should.

The push the other way follows the direction, as the push toward the truth side did. On our heads, his direction pushed back gives the mirror image too (−0.74; dark +0.71 and deception +0.50, above every random direction); our direction on his heads gives nothing (+0.02). The two crossed rows were run after the first two, with their own predictions written down first.

Caveats

- One author and one labeler: Claude wrote every scenario and read every reply (Ryan ruled the 4 unclear ones). A small batch written by someone else would test whether that matters.
- bfloat16 on one card, not his float32; the box cannot hold float32. Lie counts move by about one lie between runs (rounding steps of 0.18 bits); compare counts within one script.
- Small held-out sets: 21 lies (12 "Yes", 9 "No"), so single shares carry wide margins (about ±20 points).
- Section 2's random baselines come from the persona runs; his own test script has none for "his heads on our lies".
- Our decisions are all Yes/No at the first word; about half of his part on other words or later in the reply.

Next

The open question is which part of his data does it: owning up in sentences, or explaining at all. Ryan's hypothesis: “It seems easy to separate a plain yes-or-no lie from personality, but much harder to separate a lie you have to explain.” The next test builds directions from new scenarios where the truthful reply and the lie both explain, and from truthful replies alone, with no lie in them.

Method: Sauers' own scripts (commit 6c65875), unmodified, with our data added as a new item family; every number above is a measured value, against random directions of the same size where noted. The scenarios, labels, scripts and raw outputs are kept with the project by Ryan Jelinek.